

Werner Bisschoff

ENTERPRISE SYSTEMS & AUTOMATION ARCHITECT

Cape Town, Western Cape, South Africa

✉ werner@bisschoff.dev | ☎ 071 826 2066 | 🌐 werner.bisschoff.dev | 📱 [wbisschoff13](https://www.linkedin.com/in/wbisschoff13)

Summary

Strategic Enterprise Software Architect with 5+ years of experience engineering robust business automation platforms, custom ERP extensions, and scalable API layers. Expert in the Frappe/ERPNext ecosystem, asynchronous workflow systems, and cross-platform mobile application deployment (Expo/React Native). Proven track record of translating complex real-world operations into scalable, type-safe data schemas governed by strict Spec-Driven Development (SDD) paradigms.

Experience

Divergent Tabletop

Cape Town

PRINCIPAL SYSTEMS ARCHITECT

Jul 2025 – Present

- Engineered and hosted an enterprise-grade documentation ecosystem
- Implemented a single-database multi-tenant separation engine using PostgreSQL Row-Level Security (RLS) and transaction-scoped local parameters (**SET LOCAL**), enforcing strict data silos across isolated groups.
- Designed an atomic asynchronous flush strategy using in-memory GenServer storage buffers, reducing database persistent write transactions by 85% during periods of high user interaction.
- Utilized **Ecto.Multi** atomic chains and **Oban** transactional background workers to ensure multi-stage backend edits (e.g., structural document renaming and incoming backlink indexing) execute completely or roll back safely.
- Applied Universal Design and Executive Functioning support models to product UX/UI; established highly structured, low-context asynchronous operational playbooks that minimized cognitive friction for contributors.

FARO Africa

Cape Town

FULL-STACK ERPNext & AUTOMATION ENGINEER

Aug 2024 – Nov 2025

- Refactored and extended ERPNext via Python and JavaScript server hooks, optimizing pricing matrices and business intelligence reporting.
- Architected migration of internal Retool workflows to React Native/Expo mobile ecosystem, reducing vendor licensing overhead.
- Built custom NFC scanner communication utilities within Expo applications for instantaneous physical inventory syncs.
- Provisioned AWS infrastructure with Pulumi, deploying Inngest event-driven functions and PayloadCMS.
- Diagnosed and resolved issues in a large existing ERPNext installation; introduced LLM-assisted debugging workflows.

Ingenics Digital GmbH (through ViVa Outsourcing)

Remote Work

ENTERPRISE INFRASTRUCTURE & TOOLING ENGINEER

Mar 2023 – May 2024

- Built modular Python utility structures and automation scripts for internal configuration schema validation.
- Designed data handling pipelines ensuring data transfer consistency across runtime platforms using declarative specifications.
- Developed Python hardware mocks enabling faster development cycles with fewer hardware dependencies.

North-West University

Potchefstroom

JUNIOR LECTURER

Feb 2020 – Dec 2020

- Lectured Python and C++ programming for Introduction to Programming for first year IT students.

Education

North-West University

Potchefstroom

B.ENG. COMPUTER AND ELECTRONIC ENGINEERING

2020

- Focus on embedded systems, software engineering, and electronic design
- Developed data analysis pipeline with Python/Pandas (linear regression, ML); built NFC payment emulator (Kotlin/Android)

Technical Application Scope

Primary	Python ERPNext/Frappe Framework Expo (React Native) Inngest
Secondary	AWS Pulumi (IaC) PostgreSQL (RLS) Docker
Foundational	SQL Object-Oriented Design Spec-Driven Development
Other	C++ FreeRTOS (Low-level integration support)